SUPERSEDING
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MILITARY SPECIFICATION

COMPRESSOR UNITS, AIR/GAS, GENERAL REQUIREMENTS FOR

1. SCOPE

1.1 Scope. This specification covers the general requirements for the design and construction of air/gas compressor units (see 6.4.1). The detail requirements for a particular air compressor unit shall be as specified in the individual equipment specification for that particular air compressor unit (see 6.2).

1.2 Classification. Compressor units shall be of the following types and classes, as specified (see 6.2):

Type I Air Compressor Unit - Portable Construction

Type II Air Compressor Unit - Aircraft and Missile Support - Low Pressure

- Class 1 Gasoline engine, trailer-mounted
- Class 2 Electric motor, trailer-mounted
- Class 3 Electric motor, skid-mounted

Type III Air Compressor Unit - Aircraft and Missile Support - High Pressure

- Class 1 Gasoline Engine, trailer-mounted
- Class 2 Electric motor, trailer-mounted
- Class 3 Electric motor, skid-mounted
- Class 4 as specified

Type IV Air Compressor Unit - Base-Mounted Stationary

Type V Air Compressor Unit - Special Purpose

- Class 1 Gasoline engine, trailer-mounted
- Class 2 Electric motor, trailer-mounted
- Class 3 Electric motor, skid-mounted
- Class 4 as specified

Type VI Air Compressor Unit - Paint Spray and Garage

2. APPLICABLE DOCUMENTS

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2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

CC-M-641	Motor, Alternating Current (Integral Horsepower)
VV-F-800	Fuel Oil, Diesel

Military

MIL-V-173	Varnish, Moisture-And-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-L-2104	Lubricating Oil, Internal-Combustion Engine, Heavy-Duty
MIL-G-3056	Gasoline, Automotive, Combat
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 volts)
MIL-F-3541	Fittings, Lubrication
MIL-C-3600	Compressors, Air and Gas (Except Oxygen) Packaging of
MIL-H-4607	Heater, Engine and Shelter, Ground, Type H-1
MIL-C-4952	Cartridge, Oxygen Purifier, Type MA-1
MIL-A-5070	Adapter, Hose to Tube, Pipe and Flange, Reusable, Hydraulic, Fuel and Oil Lines
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-L-6082	Lubricating Oil, Aircraft Reciprocating Engine (Piston)
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-7365	Hose, Air Duct, for Ground Heaters
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Specification for
MIL-G-8402	Gages, Pressure, Dial Indicating, General Specification for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-L-9000	Lubricating Oil, Internal Combustion Engine, Diesel
MIL-L-10295	Lubricating Oil, Internal-Combustion Engine, Sub-Zero
MIL-B-11188	Batteries, Storage, Lead-Acid
MIL-E-11275	Engines, Gasoline, Industrial Type, General Specification for

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MIL-E-11276	Engines, Diesel, Industrial, Medium and High Speed, General Specification for
MIL-L-17331	Lubricating Oil, Steam Turbine (Noncorrosive)
MIL-C-17596	Compressors, Reciprocating, or Rotary Power Driven (E.M.D), Air, Base Mounted, 10 HP to 300 HP
MIL-L-23699	Lubricating Oil, Synthetic Base
MIL-C-26058	Cartridge, Gas Purifier, Extremely Low Dewpoint, Type MA-2
MIL-L-26087	Lubricating Oil, Reciprocating Compressor, Ground Support
MIL-H-27462	Hose Assembly, Pneumatic, High Pressure (8500 PSI)
MIL-C-83959	Cylinder, Air Purifier

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-431	Compressors: Construction, Paint Spray and Garage; Classification of Types, Classes, Styles, Groups, and Capacities
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-471	Maintainability Demonstration
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipment)
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-882	System Safety Program, for Systems and Associated Subsystems and Equipment: Requirements for
MS33586	Dissimilar, Metals

Air Force - Navy Aeronautical

AN6235	Filter Element - Hydraulic Replaceable Micronic Line Type

DRAWINGS

Air Force

69E35397	Chuck Assembly - High Pressure Air
50C24046	Coupling - Male, 6 Inch Air Duct
50C24033	Coupling - Male, 12 Inch Air Duct

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME Power Test Codes

Auxiliary Material; General Instruction and Internal-Combustion Engines Supplements on Instruments and Apparatus

Part 2, Pressure Measurement; Chapter 6
Part 3, Temperature Measurement; Chapter 6
Part 4, Head Measuring Apparatus
Part 5, Measurements of Quality of Materials
Part 6, Electrical Measurements
Displacement Compressors and Blowers
Unfired Pressure Vessels Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York 17, NY 10017.)

COMPRESSED GAS ASSOCIATION

Safety Device Standards
Pamphlet S-1 Safety Relief Device Standard

(Application for copies should be addressed to the Compressed Gas Association, 11 West 42nd Street, New York, NY 10036.)

AMERICAN STANDARDS ASSOCIATION

Standard Z24.3 Sound Level Meters
Standard Z24.10 Octave-Band

(Application for copies should be addressed to the American Standards Association, 70 East 45th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION

Pamphlet 70 National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

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DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) with Supplemental Revisions

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

NATIONAL BUREAU OF STANDARDS

Handbook H 28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D. C, 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Fungus resistance. Materials not naturally resistant to the effects of fungi and moisture shall be treated with type I varnish conforming to MIL-V-173.

3.3.2 Dissimilar metals. The use of dissimilar metals, as defined by MIL-STD-889, in direct contact with each other shall be prohibited. Metal plating or spraying of dissimilar base metals to provide suitable mating surfaces, or separation by suitable insulating materials, shall be permissible.

3.3.3 Corrosion resistance. Materials selected shall be corrosion resistant or shall be protected by plating, painting, or other surface treatment to resist corrosion.

3.3.4 Tubing. Tubing of ¼-inch outside diameter and larger which is used at pressures above 500 psi, shall be seamless, corrosion-resistant steel. All other tubing may be seamless copper.

3.3.5 Tube fittings, valves and components. All tube fittings, valves and components shall be fabricated from corrosion-resistant materials not adversely affected by extended contact with fluids such as water or corrosion cleaning solvents (trichloroethylene).

3.3.6 Threads. All screw and pipe threads shall conform to Handbook H 28. Threaded components designed for a working pressure in excess of 100 psi shall have established torque limits to prevent overstressing.

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3.4 Design and construction. Design and construction shall be as stated herein and in the detailed specification for each compressor unit type.

3.4.1 Safety. Provisions for the safety of personnel shall be incorporated to the maximum extent possible with regard to anticipated operating conditions and the capability of operating personnel. The unit shall be free of category III and category IV hazards as determined in accordance with MIL-STD-882. Equipment must be consistent with applicable "OSHA" Standards.

3.4.1.1 Guards. Guards shall be provided for moving parts such as belts, gears, and linkages to protect the operator during normal operation and guards or enclosures shall be provided for electrical portions of the compressor. High-temperature parts shall be guarded or so located that contact will not occur during normal operation. The guards shall be removable for access to the guarded parts.

3.4.1.2 Sharp edges. Sharp edges, projections, and hinged devices with hazardous characteristics are prohibited.

3.4.2 Reliability. The air compressor units shall have at 0.9 confidence the minimum reliability, for a mission time of 1000 hours and a minimum mean-time-between-failure (MTBF) of 500 hours. A reliability program shall be as specified in MIL-STD-785. Reliability testing and demonstration may be included in the preproduction test, para 4.4.

3.4.3 Maintainability. Maintainability program requirements shall be as specified in MIL-STD-470 and shall be demonstrated as required by MIL-STD-471.

3.4.3.1 Operating clearances. Maintainability provisions shall incorporate proper clearances for equipment operation, minor maintenance, and routine servicing at low ambient temperatures by personnel handicapped by wearing heavy gloves and mittens and bulky clothing.

3.4.3.2 Fastening devices. All screws, pins, belts, and similar parts shall be installed in such a manner as to prevent loss of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.4.3.3 Cover plate fasteners. Covers or plates that must be removed for component replacement or adjustments shall be equipped with substantial quick disconnect fastenings.

3.4.3.4 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. The mounting shall be so designed that the item can only be installed in its proper operating position.

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3.4.4 Pressure relief devices. Pressure relief devices conforming to the Compressed Gas Association Safety Device Standards or the ASME Unfired Pressure Vessels Code shall be provided to protect against excessive pressure in any component of the compressor unit. Provisions shall be incorporated, including consideration as to direction, to permit discharge of pressure without danger to operating or maintenance personnel. Rupture disc relief devices shall be designed to use commercially available rupture discs.

3.4.5 Safety factor. All pressurized components shall be fabricated in accordance with good engineering practice with not less than a 4 to 1 safety factor based on the minimum tensile strength of the material as tabulated in the ASME Unfired Pressure Vessels Code.

3.4.6 Compressor. The compressor shall be as specified herein for each specific type and class of compressor unit.

3.4.6.1 Number of stages. The number of stages for a reciprocating compressor shall depend upon rated pressure (see 6.4.2). Maximum compression ratio permitted shall be 4.5 to 1 for continuous operation.

Typical Pressure Range (psi gage)	Minimum Number of Stages
0 to 50	1
51 to 250	2
251 to 1,000	3
1,001 to 4,000	4
4,001 up	as specified

3.4.6.2 Capacity and pressure regulation. A compressor powered by a gasoline engine shall be provided with a continuous-engine-running type unloader, and a compressor powered by an electric motor shall be provided with a combination constant-speed unloader and a start-stop control. Simple selector controls shall be provided to permit the desired operation. The range between loading and unloading or stop-start operation shall be at the option of the contractor, and the loading-unloading and start-stop pressure range shall be adjustable with common handtools. When the compressor unit is shut down, the unloader on electric-motor-driven and gasoline-engine-driven compressor units shall automatically unload the compressor to decrease the starting load on the drive mechanism.

3.4.6.3 Pressure relief devices. Intercoolers when required shall be equipped with adjustable, spring-loaded type pressure relief valves designed and adjusted to operate at a pressure that will protect the compressor. The receivers shall have a rupture-disc type pressure relief device in addition to an adjustable, spring-loaded relief valve. The relief valve shall be set at a lower pressure than the rupture-disc device. Relief valves shall be set at 10 per cent above the maximum working pressure.

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3.4.8.1.2 Military standard engines. When a DOD military standard engine is used, the total maximum power required including all drive and accessory loss measured at the engine flywheel shall not exceed the continuous rated BHP specified for any particular crankshaft speed. Military standard engines shall operate at not less than 2,600 rpm nor more than 3,800 rpm during normal operation.

3.4.8.1.3 Diesel engine. When a diesel engine is used it shall be a single, lightweight, multicylinder, liquid-cooled, 2- or 4- stroke cycle diesel engine conforming to MIL-E-11276. The engine size shall be such that the total driven load (including losses in the power take off and power transmission system) during operation of the compressor under any of the conditions specified herein will not exceed the continuous (24 hours per day) output horsepower rating of the engine.

3.4.8.1.4 Cylinder head temperature. Air-cooled engine installations shall be such that the cylinder head temperature, as measured by a gasket-type thermocouple installed under any sparkplug, will not exceed 525° F when the power unit is used as specified herein.

3.4.8.1.5 Lubricating oil temperatures. The engine installation shall be such that the lubricating oil temperature, as measured by a probe-type element inserted into the oil sump, will not exceed 225° F when the engine is used as specified herein.

3.4.8.1.6 Fuels and lubricants. The engine power source shall be designed to operate as specified herein when supplied with grade 80/87 gasoline conforming to MIL-G-5572, type I or II gasoline conforming to MIL-G-3056, or diesel fuel oil in accordance with VV-F-800. Lubricating oils for the engines shall be in accordance with MIL-L-2104, MIL-L-6082, or MIL-L-9000.

3.4.8.1.7 Clutch. A manually controllable clutch shall be provided on types III and V compressor units to permit disengagement of the engine from the compressor for starting and operating the engine without load. The clutch shall be designed to withstand repeated engagement without damage when the compressor is loaded, or a means shall be provided to prevent engagement when the compressor is loaded.

3.4.8.1.8 Variable speed control. A variable speed control for all engines shall be provided to permit manual control of the engine speed. The engine speed shall be controllable from idling to rated capacity power.

3.4.8.1.9 Engine accessories. The following minimum engine accessories shall be provided:

a. An oil pressure gage (for pressure lubrication systems), a cylinder head high temperature safety cut off switch, a governor, and a fuel gage.

b. A tachometer shall be provided for diesel engines.

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e. In addition to those specified herein, all accessories that are necessary for safe and efficient operation and control of the engine shall be provided.

3.4.8.1.10 Fuel tank. A fuel tank designed for use with the fuels specified in 3.4.8.1.6 shall be provided. The installation of the fuel tank shall be made in such a manner that the tank can be easily removed for repair or replacement, and the connections necessary for securing the tank to the compressor unit shall be made in a manner that will prevent fatigue failure in the tank or in the connections as a result of vibrations induced by operation of the compressor unit. The fuel tank shall have sufficient capacity to operate the compressor at rated output for 8 continuous hours. The filler neck shall have a minimum inside diameter of 1-3/4 inches for fuel tanks of 25-gallon capacity or less, and 2-½ inches for larger fuel tanks. The filler shall be located for easy external access and shall not permit spilled fuel to collect within the compressor access unit. The tank shall be vented to the outside of the compressor unit. The tank shall be equipped with a drain port. The area around the drain port shall be dimpled or lower than the bottom of the tank to facilitate complete and easy drainage of fuel.

3.4.8.1.10.1 Fuel level indicator. A fuel level indicator shall be mounted adjacent to the fuel filler. The indicator shall be accurate to within 5 percent of the full-scale range when the fuel tank is being serviced. The tank shall be so baffled that the rate of tank filling will have no adverse effects on the accuracy of the indicator.

3.4.8.1.11 Electrical starting system. Gasoline-engine-powered compressors shall be equipped with an electrical starting system. The electrical starting system shall include an electrical cranking motor, generator/alternator, battery, voltage regulator, ignition switch, and other related components. Starting system voltage shall be as specified in the detail specification.

3.4.8.1.11.1 Storage batteries. Storage batteries shall be in accordance with MIL-B-11188. The batteries for a compressor may be composed of either all 12-V or 24-V size, as applicable.

3.4.8.1.11.1.1 Battery box. Compressor units requiring batteries shall be provided with a corrosion and acid-resistant box incorporating provisions for gas venting, drainage, winterization when required, rigid support, and accessibility for examination of battery terminals and electrolyte level, and liquid replenishment of all cells. The box shall be so located that the battery can be easily removed.

3.4.8.1.11.1.2 Terminals. Standard noncorrosive, screw type terminals shall be provided on the battery end of the battery cables.

3.4.8.1.11.2 Starter motor. The electric starter motor shall be designed to repeatedly crank the engine at sufficient starting speed when the engine is serviced with lubricants as specified herein and with the engine at a temperature not exceeding 0° F. When winterization is required the drive mechanism for engaging the starter motor with the engine flywheel shall be designed to operate and release without the application of heat down to an ambient temperature of -65° F.

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3.4.8.1.11.3 Charging generator/alternator. The battery-charging generator/alternator shall be belt driven with a charging rate sufficient for the services required. Generators/alternators shall be in accordance with MIL-E-11275 or MIL-E-11276, as applicable, and generators shall be of the two-brush design. Regulators, including the reverse-current relay and voltage current regulator relay units, shall be provided and shall be adjustable to provide sufficient range of circuit voltage.

3.4.8.1.11.4 Ammeter. An ammeter shall be provided to indicate charge and discharge of the battery current.

3.4.8.1.12 Exhaust system. A multiple-baffled muffler or mufflers shall be provided and shall be capable of handling the exhaust gasses without causing more than 2 inches Hg backpressure in the engine-exhaust-manifold system under any operating condition. Exhaust pipes shall be of such length to insure discharge of exhaust gasses to a safe location with respect to the operator and unit including raised drawbar or any other compressor enclosure surface.

3.4.8.2 Electric motor requirements. Electric motors shall be of the squirrel-cage induction type in accordance with CC-M-641. The ambient temperature referenced shall be 122° F, and the motors shall be capable of operation under the conditions specified herein. The continuous load on the motor shall not exceed the nameplate rating for continuous operation. When explosion-proof motors are specified, the motors shall be capable of operating under all specified conditions without exceeding the nameplate rating.

3.4.8.2.1 Motors rated 1 hp and under. Motors rated 1 hp and under shall be designed to operate on 120/230-V (1-phase, 3-wire), 60-hertz, a-c power. Either manual or magnetic-type across-the-line starting may be provided.

3.4.8.2.2 Motors rated 1 to 20 hp. Motors rated above 1 hp shall operate on 230/460-V (3-phase, 4-wire), 60-hertz, a-c power. Magnetic connectors for across-the-line starting shall be furnished for motors up to 20 hp and shall be capable of operation on 230- or 460-V power. Alternate thermal protectors not used in initial wiring shall be securely fastened inside the starter box to prevent loss. The motor starter shall be wired as shown in figure 1.

3.4.8.2.3 Motors rated above 20 hp. Motors rated above 20 hp shall have reduced voltage, partial winging, or other means to limit starting current to 200 percent of the normal full-load value. Equipment shall be connected for 460-V operation. Control circuits shall be nominal 110 V and components shall be equipped with fused or automatic breaker devices to protect the circuits from overload conditions.

3.4.8.2.4 Reverse rotation protection. Three-phase electric motors on compressor units that will be damaged by reverse rotation shall be provided with equipment to prevent rotation of the compressor in the wrong direction in event the phase sequence is wrong. Means shall be provided to reverse the phase sequence in one simple operation without resorting to rewiring of plugs or switches, on all three-phase electric motors.

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3.4.8.2.5 Electrical power. The compressor unit shall be designed and constructed to perform as specified herein when supplied with 115 V ± 10 percent, 230 V ± 10 percent or 460 V ± 10 percent, at 60 ± 3 hertz ac, as applicable.

3.4.8.2.6 Motor protection. Overload thermal protection devices of applicable power-handling capacity shall be provided for each electric motor.

3.4.8.2.7 Electric cables. Electric cables shall conform to MIL-C-3432, Class HD or HD, flexibility E. Strain relievers shall be provided on the cables.

3.4.8.2.7.1 Input cables (single-phase motors). Single-phase electric motors shall be provided with a permanently attached 6-foot, 3-conductor cable. The third conductor shall serve as a ground for the compressor unit, and the compressor unit shall be capable of operation without the third conductor connected.

3.4.8.2.7.2 Input cables (3-phase motors). The input cable for 3-phase electric motors shall be a 50-foot, 6-conductor cable and shall be connected as shown in figure 1. Cable leads L_1, L_2, L_3 shall be sized as required. Ground lead L_0 shall not be smaller than size 8. Control leads C1 and C2 shall not be smaller than size 12. For power conductors (L_1, L_2, and L_3) smaller than size 8, the power and ground conductors shall be the same size, but not smaller than size 12. The free end of the cables shall have the outer sheath removed for a distance of 12 inches to permit the attachment of a plug by the using activity.

3.4.8.2.7.3 Rack or reel. A rack or reel capable of containing the electric power cable shall be provided. It shall provide a simple, hand-operable means of positively holding the free end of the cable to prevent unwinding during storage and transportation. It shall be permanently attached to the compressor unit in a conveniently accessible place.

3.4.8.3 Shutoff and control valves. Manually operated shutoff and control valves shall be provided as required. The valves shall be designed and constructed to provide positive shutoff when closed and minimum restriction to flow when open. The valve shall be so designed that the bonnet or stem packing nut will not come loose while turning the valve handle while under pressure. The valves shall be designed and mounted to permit replacement of the valve seats and seals without removing the valve from the compressor or the valves shall be located to permit removal and replacement without removing any other item on the air compressor unit.

3.4.8.3.1 Valve handles. Each manually operated shutoff or control valve shall be provided with a tee, cross, or other easily turned type handle with a diameter of not less than 3 inches per inch of nominal size, but in no case less than 1 inch.

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3.4.8.3.2 Valve leakage. The valve shall show no indication of leakage with the outlet plugged or capped, with the maximum working pressure of the component in which it will be installed applied to the inlet, and with the valve either open or closed. The valve shall not leak more than 2.5 cubic inches of free gas per hour per half-inch or fraction thereof of nominal size from the maximum working pressure of the component in which it will be installed to a downstream pressure of atmospheric under the specified environmental conditions and when closed with a torque of 60 ± 5 pound-inches per inch of nominal size. In addition, the valve shall be capable of completing not less than 2,000 cycles of operation (see 6.4.8), including 1,000 cycles with both inlet and outlet subjected to maximum working pressure and 1,000 cycles exhausting from maximum working pressure to atmospheric pressure, without exceeding the specified leakage limit.

3.4.8.3.3 Valve overtorque performance. The valve shall not require more than a closing torque of 60 ±5 pound-inches per inch of nominal size to seal against a differential pressure equal to the maximum working pressure of the component in which the valve will be installed when used under the specified environmental conditions. In addition, the valve shall be designed and constructed to withstand a closing torque of 300 ± 10 pound-inches per inch of nominal size under any of the specified environmental conditions without parts failure or leakage in excess of that permitted by 3.4.8.3.2 when subsequently closed with a torque of 60 ± 5 pound-inches per inch of nominal size.

3.4.8.3.4 Valve mounting. All manually controlled valves shall be so mounted that they can be readily opened or closed by an operator standing at the equipment control panel. All valves shall be so mounted to supporting members or large size piping so that the forces applied to the control handles in opening or closing the valves will not damage the equipment piping.

3.4.8.4 Hourmeter. An hourmeter shall be furnished and mounted on the control panel of all compressor units that include a dehydration system to provide clean, dry air or when specified in the detail specification (see 6.2). The hourmeter shall record total operating time of the compressor unit and shall have a range from 0 to 9,999 hours and accuracy of 1 minute per hour.

3.4.9 Lubrication system, military lubricants shall be used. Sufficient lubrication shall be provided to all moving parts of the compressor unit. The lubricating system shall include an oil level indicator or bayonet gage, a drain, and, when applicable, a crankcase ventilator. The lubrication system shall be designed to operate under the range of conditions specified herein. Compressor unit lubrication fittings shall conform to MIL-F-3541, types I, II, and III, and shall be located in accessible, protected positions.

3.4.9.1 Parts subject to wear. Provisions shall be made for lubrication of all parts subject to wear. Wherever practicable, permanently lubricated components or units not requiring lubrication shall be used. Grease and oil seals shall be designed and located to provide maximum accessibility for inspection and replacement.

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3.4.10 Mounting provisions. The compressor unit shall be mounted on a trailer assembly, a skid-mounted assembly, or a stationary base (see 1.2).

3.4.10.1 Trailer mounting. The compressor shall be mounted on either a 2- or 4-wheel trailer conforming to the requirements of MIL-M-8090 as specified in the detail specification. Trailers may be a separate component or may be formed by attaching the running gear, towbar, and necessary trailer components directly to the compressor unit.

3.4.10.1.1 Four-wheel running gear.

3.4.10.1.1.1 Towbar latch. Compressor units shall be provided with a positive-type latch to hold the towbar in the up, vertical position at any position of the front wheels. The towbar shall have a stop in the up position to prevent contact with the frame or enclosure.

3.4.10.1.1.2 Bushings and lubrication fittings. All pivot points associated with the steering mechanism of four-wheel running gear assemblies shall be equipped with bushings and lubrication fittings.

3.4.10.1.2 Two-wheel running gear. Two-wheel-mounted compressors shall have a two-wheel running gear and a full-swivel landing wheel.

3.4.10.1.3 Tilted operation. The compressor unit shall be designed to deliver full-rated capacity at rated pressure when tilted in any position at an angle of 15° from the horizontal.

3.4.10.1.4 Center of gravity. The center of gravity of all trailer-mounted compressor units shall be such that the compressor unit will not overturn when the axles form a 45° angle with the horizontal. The use of excessive tread width will not be permitted.

3.4.10.2 Skid mounting. Skid-mounted compressor units shall include a heavy, welded, structural-steel frame consisting of two skids with a closure between them. The skids shall be sufficiently far apart and the center-of-gravity of the unit such that no tipping of the unit will occur when it is placed in any direction on a 15° slope. All members shall be so braced and gusseted that there will be no objectionable sagging or distortion of the unit. Both ends of the frame shall be reinforced, and each end shall be provided with rings or other means to permit pulling the unit. The bottom plates of the skids forming the runners shall be renewable and shall be curved or turned up at the ends to prevent snubbing when the compressor unit is pulled over rough ground. Skids of boxed or closed construction shall be provided with drains.

3.4.10.3 Stationary mounting. The compressor unit shall be designed for mounting on a single, permanent, concrete slab base. Multilevel slab-type mounting will not be permitted.

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3.4.11 Enclosure. A weatherproofed metal enclosure shall be provided for the compressor unit when required. Controls shall be incorporated into the enclosure to maintain normal operating temperature around the power source, compressor, and any components susceptible to freezing. Provisions shall be incorporated to provide ventilation for the engine and compressor to prevent overheating.

3.4.11.1 Control panel. A control panel shall be provided upon which the compressor unit controls, power unit controls, receiver valves, and gages shall be mounted. The control panel shall be located within the enclosure, shock-mounted, and accessible through a large door, and with the control panel door in the opened-and-locked position, the enclosure shall not be vented.

3.4.11.2 Lighting of enclosed compressor units. Lighting provisions shall be incorporated on the control panel to permit night operation. The lights shall be shielded from above to direct minimum light away from the control panel. The lights shall operate from the electrical system of the compressor unit and shall be controlled by a switch on the control panel.

3.4.11.3 Winterization provisions. When winterization is required, the enclosure of the compressor unit shall be designed for connectinn to a type H-1 heater conforming to MIL-H-4607. Provision shall be included for attachment one 12-inch diameter duct or three 6-inch diameter duct connection openings. Insofar as applicable, duct connections and covers shall be in accordance with MIL-H-7365. The coupling at the compressor unit shall be a male coupling in accordance with Drawing 50C24046 or 50C24033. Dust covers shall be securely fastened to the enclosure by a chain or other means to prevent loss. If necessary, heater ducting shall be provided within the compressor unit enclosure to direct the heated air to thoroughly free all critical moving parts that may be restricted due to congealed oil or ice. Duct attachments shall not project beyond the enclosure of the compressor unit.

3.4.11.4 Doors and covers. Doors and hinged covers shall be provided for convenient access to the power drive unit, compressor, and control equipment. They shall be reinforced as required to provide rigidity. The doors and covers shall be furnished with positive flush latches or fasteners to hold them securely in the closed position and positive locks to hold them in the open position.

3.4.11.5 Roof. The roof of the enclosure shall be readily removable so that installed components can be vertically removed or replaced. The roof fastening device shall be positive to preclude loss of the roof when traveling at highway speeds.

3.4.11.6 Winds. The enclosure shall be designed to withstand winds up to 70 mph without damage. Doors and hinged covers that are normally locked in the open position for operation shall also withstand the 70 mph winds.

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3.4.11.7 Utility compartment. A compartment shall be provided on all but type IV compressor units for the storage of inspection forms, and special tools. The compartment shall be covered to protect the contents from oil, grease, dirt, and rain. The compartment shall be a minimum size of 9 by 11-½ by 4 inches.

3.4.12 Control-panel-mounted pressure gages. The gage case shall be waterproof and shall be constructed in such a manner to preclude collection of condensate in the gage. A standard type case will be allowed with a safety release disc in the back and the gage shall be equipped with a shatterproof glass or plastic face or the standard glass face shall be protected with a plastic shield.

3.4.13 Air transportability. Types II, III, and V compressor units shall comply with MIL-A-8421.

3.4.14 Lifting provisions. Lifting provisions are applicable to the types I, II, III, V, and VI compressor units.

3.4.14.1 Lifting rings. The compressor unit shall be provided with one or more lifting rings for cable attachment and hoisting without damage to the compressor unit. The lifting ring or rings shall be so located on the compressor unit that transportation personnel can rig a safe lifting sling from common cable and spreader bar components for lifting by a single-hook, overhead crane. The lifting ring or rings shall be constructed and installed to permit lifting the compressor unit without damage. The ability of the compressor unit to withstand the lifting attachment acceleration loadings shall be demonstrated by supporting engineering data.

3.4.14.2 Lifting eyes. Lifting eyes or other hoisting attachments shall be provided on major individual compressor unit components whose weight exceeds 75 lbs. such as the compressor and power unit so that they may be readily removed from or replaced on the compressor unit chassis or base.

3.4.15 Individual type requirements.

3.4.15.1 Type I. Type I compressor units shall comply with MIL-STD-431.

3.4.15.2 Type II. Type II compressor units shall consist of the following major components:

a. Compressor
b. Power drive unit
c. Receiver
d. Trailer chassis with or without enclosure, or skid mounting without enclosure.

3.4.15.2.1 Compressor. The compressor shall be of the reciprocating or rotary type with a continuous output of 200 psi and an intermittent output of 250 psi as follows:

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3.4.15.2.1.1 Reciprocating type. The compression shall be carried out in two or more separate and distinct stages. Intercoolers shall be provided to cool the air between stages, and an aftercooler shall be provided to cool the air before entering the receiver. The intercoolers and aftercooler shall be designed to cool the air to within 30° F of ambient temperature, in ambient temperatures of 20° to 125° F. Intercoolers and aftercoolers shall be equipped to automatically drain off condensed moisture and oil. Traps to collect condensed moisture and oil shall have a full-load operation capacity of 2 hours.

3.4.15.2.1.2 Rotary type. The air shall be oil-cooled, and means provided to remove and recover the oil from the air. Air temperature during compression shall not exceed 250° F during operation in ambient temperatures up to 125° F. An aftercooler shall be provided to cool the air after compression and before entering the receiver to within 30° F of ambient temperature in an ambient temperature range of 20° to 125° F. Provisions shall be made in the aftercooler installation to automatically drain all moisture and oil into a trap with a full-load operation capacity of 2 hours.

3.4.15.2.2 Power unit. Type II, class I compressor units shall be powered by gasoline engines. Type II, classes 2 and 3 compressor units shall be powered by explosion-proof electric motors.

3.4.15.2.3 Outlet pressure regulator. The compressor unit shall be provided with a pressure regulator designed to regulate the outlet pressure within the range of from 5 to 250 psi. The regulator shall regulate the outlet pressure at the desired pressure within ± 5 percent. The outlet hose connection shall be on the outlet side of the pressure regulator.

3.4.15.2.4 Air hose. A 50-foot length of 5/16-inch nominal inside diameter air hose in accordance with MIL-H-8795 shall be provided. The hose shall be equipped with end fittings in accordance with MIL-A-5070. A screw-on-type air chuck in accordance with Drawing 69E35397-5 shall be provided and attached to the hose with a suitable fitting. A rack shall be provided on the compressor unit upon which the hose may be wound.

3.4.15.2.5 Receiver. The receiver shall be equipped with pressure relief and drain valves. The drain valves shall be so designed and located that they will not be damaged or exposed to damage when the compressor unit is being towed over obstructions. A siphon-type drain shall not be used. The drain valves shall drain off all condensed moisture that collects in the receiver.

3.4.15.2.6 Gages. Gages shall be provided to indicate the pressure in the receiver and at the outlet of the pressure regulator. The gages shall cover a range of from 0 to 300 psi, with 5-psi or smaller increments, and shall be permanently redlined on the gage face at 250 psi. The gages shall have a dial diameter of not less than 2½ inches. They shall be accurate to within ± 2 percent of rated maximum pressure.

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3.4.15.2.7 Air filter. An oil-bath or dry cartridge type air filter shall be provided to filter the air being inducted into the first stage of the compressor. The filter shall be designed to remove harmful dust and sand particles from the intake air to the maximum practicable extent.

3.4.15.2.8 Unloader. The range between loading and unloading shall not exceed 50 psi. The unloader pressure shall be adjustable from 100 to 250 psi. The compressor unit shall be set to unload at 200 psi.

3.4.15.2.9 Pressure. The compressor unit shall be required to deliver the specified capacity of free air (see 6.4.3) at a pressure of 200 psi for continuous operation. The compressor unit shall be designed to deliver the specified capacity of free air at a pressure of 250 psi for intermittent operation.

3.4.15.2.10 Endurance. The compressor unit shall be designed for continuous operation under the conditions specified herein. It shall be capable of 1,000 hours of operation between major overhauls.

3.4.15.2.11 Valves. Valves shall be in accordance with 3.4.8.3.

3.4.15.3 Type III compressor units. Type III, classes 1, 2, and 3 compressor units shall be in accordance with the following requirements and shall consist of the following major components:

a. Compressor
b. Dehydrator and filter assembly
c. Trailer or skid mounting as specified
d. Controls

3.4.15.3.1 Compressor. The compressor shall be of the reciprocating, air-cooled type equipped with intercoolers and an aftercooler. The intercoolers when required shall cool the air to within 30° F ambient temperature for ambients of 10° to 125° F. Each intercooler moisture separator when required and the aftercooler moisture separator shall be designed to automatically drain off all condensed moisture and oil upon shutdown of the unit. The intercooler and aftercooler moisture separators shall also be provided with a manual drain for draining off all condensed moisture and oil. The aftercooler shall cool the air to within 25° F of ambient temperature at the inlet to dehydration equipment for ambients of 10° to 125° F. In its passage to the servicing valve, the air shall not be reheated above the outlet temperature from the dehydration equipment.

3.4.15.3.1.1 Separators. When mechanical separators are used, they shall be constructed from materials that will not deteriorate or break up during use. If periodic servicing is required, the units shall be designed to permit servicing with a minimum of disassembly.

3.4.15.3.1.2 Compressor oil filter. Type III compressors shall be equipped with lubrication oil filters. If full-flow, cartridge-type replaceable filters are provided, they shall be in accordance with MIL-E-20707. If permanent filters are provided, they shall be of a type that can be easily removed and cleaned.

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3.4.15.3.1.3 Crankcase vent. For type II air compressors the crankcase shall be vented through an oil separator into the inlet to the first stage cylinder. The separator shall separate the oil and return the oil to the crankcase.

3.4.15.3.2 Control panel. A control panel shall be provided upon which the compressor unit controls, power unit controls, and receiver valves and gages shall be mounted. When the compressor unit requires an enclosure the control panel shall be located within the enclosure, shock-mounted, and accessible through a large door. With the control panel door in the opened-and-locked position, the enclosure shall not be vented. The control panel shall include the following gages and controls:

a. Pressure gages - intercoolers and aftercooler pressure (psi)
b. Pressure gage - compressor lubricating oil pressure (psi)
c. Servicing gage - high pressure
d. Servicing gage - low pressure
e. All engine controls and gages or all motor starting and running equipment.
f. Hourmeter, when applicable (see 3.4.8.4).

3.4.15.3.3 Dehydrator and filter assembly. Provisions shall be furnished to dry and clean the outlet air.

3.4.15.3.3.1 Type MA-2 gas purifier cartridges conforming to MIL-C-26058 shall be used for dehydration. Cylinders conforming to MIL-C-83959 shall be used to hold the cartridges when the operating pressures do not exceed 4,000 psi. Two cylinders shall be mounted in series with the gas entering the bottom of each cylinder. The cylinders shall be mounted in the upright, vertical position, measured when the compressor unit is in the normal, level position.

3.4.15.3.3.2 Backpressure control valve. A backpressure control valve shall be provided at the outlet of the dehydrator assembly to prevent passage of air through the dehydrator assembly below 3,000 psi unless otherwise specified in the detail specification.

3.4.15.3.3.3 Filter assembly. A filter assembly shall be provided after the dehydrator assembly to remove particles down to 10 microns. The filter shall be readily removable for cleaning or element replacement. When replaceable elements are used, they shall conform to filter element replacements in accordance with AN6235, if practicable.

3.4.15.3.3.4 Bleed valve. A bleed valve shall be provided downstream of the final compression stage and in advance of the 3000-psi back pressure control valve to permit bleeding the line pressure before changing cartridges or filter element.

3.4.15.3.4 Air hose. For operating pressures up to and including 4,000 psi, servicing hose assemblies shall be in accordance with MIL-H-8795 and, unless otherwise specified, shall be provided with a screw-on-type air chuck in accordance with Drawing 69E35397-5. For pressures over 4,000 psi, but not exceeding 8,500 psi, servicing hose assemblies shall be in accordance with MIL-H-27462.

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3.4.15.3.4.1 Hose rack. A rack or reel shall be provided in a convenient location for storage of the servicing hose. It shall provide a simple, hand-operable means of positively holding the hose to prevent unwinding or uncoiling during storage and transportation. Disconnection of the compressor unit end of the hose shall not be allowed for normal storage. The rack or reel shall be designed to assure that the critical bend radii of the hose is not exceeded when in the stored position. On compressor units with enclosures, the hose rack shall be mounted within the enclosure.

3.4.15.3.5 Trailers. Type III, classes 1 and 2 compressor units shall be mounted on 2- or 4- wheel trailers, as specified.

3.4.15.3.6 Receiver. A receiver conforming to 3.4.15.2.5 shall be provided.

3.4.15.3.7 Continuous operation. The compressor unit shall be capable of continuous operation under the conditions specified. It shall be capable of 1,000 hours of operation between major overhauls.

3.4.15.3.8 Discharge air. Discharge air shall maintain a dew-point (see 6.4.4) of -65° F, or less, based on the frequency of cartridge changes as determined during endurance testing.

3.4.15.3.9 Low-pressure servicing gage. In addition to the high-pressure servicing gage, another gage shall be provided to indicate servicing hose pressure for low-pressure pneumatic charging operations. The gage shall cover a range of from 0 to 600 psi, with 5-psi or smaller increments. The gage shall have a dial diameter of not less than 4½ inches, and shall be accurate to within ± 2 percent of full-scale reading. The gage shall conform to MIL-G-8402 and shall withstand an overpressure of 900 psi.

3.4.15.3.10 Blowdown receiver. Provisions shall be incorporated to prevent visible condensates from blowing to the atmosphere when the compressor unit is unloaded. The condensate receiver shall be of sufficient capacity to provide for a minimum of 8 hours operation.

3.4.15.4 Type IV compressor unit. The type IV compressor unit shall be in accordance with MIL-C-17596.

3.4.15.4.1 Electric cables. Cables and plugs need not be furnished with type IV and V compressor units having electric motors of greater than 20 hp.

3.4.15.5 Type V compressor units. Type V compressor units shall be defined in the applicable specification.

3.4.15.6 Type VI compressor unit. The type VI compressor unit shall be in accordance with MIL-STD-431.

3.5 Performance.

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3.5.1 Environmental conditions.

3.5.1.1 Operation at extreme temperatures. Types II, III, and V compressor units shall be capable of starting and operating in any ambient temperature (see 6.4.5) between 0° and 125° F without the aid of external heaters plus exposure to solar radiation on the surface of 359 BTU/sq ft/hr at the 125° F ambient temperature. Oil dilution will not be acceptable.

3.5.1.1.1 Low temperature starting and operation. When equipped with winterization provisions and an enclosure (see 3.4.11.3), the compressor unit shall be capable of starting and operating at -65° F. External heaters as specified herein and lubricating oil having a viscosity not lower than SAE No. 10 wt shall be used. The compressor unit shall be capable of starting within ½ hour and, once started, shall operate satisfactorily without external heating.

3.5.1.2 Rain. When equipped with an enclosure, the compressor unit shall be capable of starting and operating during exposure to rain.

3.5.1.3 Humidity. Types II, III, and V compressor units shall not be damaged by storage in relative humidities up to and including 100 percent, including condition wherein condensation takes place in the form of water and frost.

3.5.1.4 Fungus. Types II, III, and V compressor units shall not be damaged by exposure to fungus growth such as encountered in tropical and subtropical climates.

3.5.1.5 Salt atmosphere. Types II, III, and V compressor units shall not be damaged by operation or storage in an atmosphere containing salt-laden moisture such as encountered near bodies of salt water and in transportation on shipboard.

3.5.1.6 Sand and dust. Types II, III, and V compressor units shall not be damaged by operation or storage in an atmosphere containing airborne sand and dust particles such as encountered in desert locations.

3.5.2 Storage at extreme temperatures. The compressor units shall be capable of withstanding storage at temperatures of between -80° and + 160° F without damage for periods of up to and including 72 hours.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Electromagnetic interference. The requirements of MIL-STD-461 shall apply as specified in the detail specification.

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3.8 Audio noise levels. The maximum permissible noise levels for types II, III, and IV compressors shall not exceed the following values when measured as specified in 4.7.16:

Octave Band Pressure Level (In Decibels Ref. 0.0002 Microbar)

Frequency Bands	Operator	Average at 25 Feet
35-75	106	91
75-150	96	81
150-300	88	73
300-600	85	70
600-1200	84	69
1200-2400	83	68
2400-4800	82	67
4800-9600	81	66

3.9 Finishes and protective coatings.

3.9.1 Cleaning, painting, plating, anodic films, and chemical treatments. All cleaning, painting, plating, chemical films, and chemical treatments shall be in accordance with MIL-STD-808. Parts subjected to high temperatures shall be finished in accordance with finish code FF-903.

3.10 Operational markings.

3.10.1 Certification plate. All pressure vessels shall bear a permanent certification plate or permanent stamp stating that they conform to the ASME Unfired Pressure Vessel Code. The date of initial hydrostatic test shall be permanently recorded on the certification plate.

3.10.2 Operating instructions. Brief operating instructions shall be permanently affixed on or near the control panel. The instructions shall be clear, concise, and sufficient to enable operation without damage to the compressor unit or injury to personnel. The instruction plate shall be constructed of sheet aluminum or sheet zinc of not less than 0.050 inch thickness, anodized or etched to produce raised markings with a black or dark color background, and with a raised border not less than ¼ inch wide.

3.10.2.1 Identification. All valves, gages, and indicators shall be identified by nameplates or similar identification.

3.10.2.2 Operational pressure. All hose racks or reels shall be marked by decal or stencil to indicate the maximum operational pressure anticipated for the hose stored thereon.

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3.10.3 Wiring diagram. A complete electrical wiring diagram shall be permanently affixed on or near the control panel in a readily accessible and protected location that will be within the housing assembly (if applicable) when it is closed for transport or storage. The diagram shall be of sufficient size for clear readability. It shall be constructed of sheet aluminum or sheet zinc of not less than 0.05 inch thickness, anodized or etched to produce raised markings with a black or dark color background, and with a raised border of not less than ¼ inch wide.

3.10.4 Voltage conversion instructions. Complete and readily understandable instructions for converting from one supply voltage to the other shall be either incorporated into the electrical wiring diagram or shown on a separate instruction plate adjacent to it. A separate conversion instruction plate shall be constructed of sheet aluminum or sheet zinc of not less than 0.05 inch thickness, anodized or etched to produce raised markings with a black or dark color background, and with a raised border of not less than ¼ inch wide.

3.10.4.1 Motor input voltage identification plate. A corrosion-resistant, reversible metal plate shall be provided and permanently marked with one input voltage on one side and the alternate voltage on the other side. The plate shall be located under or directly adjacent to the starter switch and so attached that it may be easily reversed to indicate the nominal voltage for which the machine is connected.

3.10.5 Airflow diagram. A schematic airflow diagram shall be attached to the inside panel or door to the control panel, along with the nomenclature of the controls and gages, and operating instructions.

3.10.6 Type of fuel and tank capacity. The type of fuel to be serviced and the capacity of the fuel tank shall be stenciled on the compressor unit adjacent to the filler opening.

3.10.7 Caution plates. Caution plates shall be attached close to the compressor unit lubrication fittings where high-pressure lubrication equipment, 1000 psi or higher, would damage grease seals or other parts.

3.10.8 Lubricating points. A bright red circle shall be painted around each lubricating point of the compressor unit.

3.10.9 Instructions for changing cartridge holders. A corrosion-resistant, metal plate containing the following instructions for changing the cartridges in the cartridge holders shall be provided in an accessible location near the dehydrator assembly. The cartridge holders shall be labeled No. 1 (cartridge holder that air first enters) and No. 2 (cartridge holder following the No. 1 holder). A blank space shall be provided on the cartridge change instructions on which the hourmeter reading can be recorded by pencil and erased. Instructions for changing cartridges shall be as follows:

a. Use oxygen type MA-2 gas purifier cartridges conforming to MIL-C-26058.

b. Change cartridges at intervals indicated below. Record hourmeter reading at time of change.

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Ambient Temp. Range °F	Total Hours of Operation Between Change
-65 to 0	50
0 to 70	20
70 to 100	15
100 to 125	7
Cartridges changed	

This space shall be of a material on which pencil markings can be placed or erased.

c. Change cartridges as follows:

(1) Discard cartridge in No. 1 (inlet) holder

(2) Transfer cartridge from No. 2 (outlet) holder to No. 1 holder

(3) Place fresh cartridge No. 2 holder.

d. Replace seals with No. (AN STD No.) O-rings when necessary.

3.10.10 Shipping data plate. A corrosion-resistant metal plate containing the length, width, height, curb weight, gross vehicle weight and the center or balance of the unit shall be provided in an accessible location.

3.11 Identification of product. The nameplate markings shall be in accordance with MIL-STD-130.

3.12 Workmanship. Attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts, and tightness of assembly screws and bolts.

3.12.1 Cleaning. The compressor unit shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.13 Government-furnished property. When specified in the contract the Government will furnish the following to the contractor, upon his request, for use in making or testing the compressor units:

Description	Quantity
Military standard engine and accessories	As specified
H-1 heater conforming to MIL-H-4607	As specified

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the compressor unit shall be classified as follows:

a. Preproduction tests ..... See 4.4

b. Acceptance tests ..... See 4.5

c. Preinstallation tests ... See 4.6

4.3 Test conditions.

4.3.1 Test codes. Tests shall be in accordance with the following applicable ASME Power Test Codes. Method of testing shall be determined, instruments selected and calibrated, and tests run in accordance with the ASME codes to result in the degree of accuracy specified herein.

a. General instructions

b. Internal-combustion engines

c. Part 2, pressure measurement; chapter 1, 2, and 4

d. Part 3, temperature measurement; chapter 6

e. Part 4, head measuring apparatus

f. Part 5, measurements of quantity of materials

g. Part 6, electrical measurements

h. Displacement compressors and blowers

i. Unfired pressure vessels code.

4.3.2 Airflow. The airflow shall be measured by means of smooth approach orifices as specified in the ASME Power Test Codes.

4.3.3 Tolerances. The capacity, pressures, temperatures, and other data required for the performance demonstration shall be accurate within $\pm$ 2 percent. Data on compressor unit operation during endurance, environmental, and individual tests shall be accurate within $\pm$ 5 percent.

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4.3.4 Fuels and lubricants. The engine shall use oils in accordance with MIL-L-2104, MIL-L-6082, and MIL-L-9000 and fuels as specified herein. The compressor shall use lubricants in accordance with MIL-L-17331, MIL-L-23699 or MIL-L-26087 as applicable. MIL-L-2104 and MIL-L-10295 may be used for compressor lubricant in the Type II compressor when supplied without a clutch.

4.3.5 Inspection, servicing, and adjustments. Planned stops shall be made at 8-hour intervals during the endurance test for inspection, servicing, and adjustments. Oil may be changed at 100-hour intervals, and added as required.

4.3.6 Test time. Endurance test time shall be credited in ½-hour intervals, except where shorter periods are a test requirement.

4.3.7 Barometric pressure. Barometric pressure shall be measured by means of a calibrated mercury or aneroid barometer and shall be corrected for temperature. When an aneroid barometer is used, a certified calibration shall be furnished, and an additional mercury barometer reading shall be supplied at weekly intervals during the testing.

4.3.8 Temperatures. Temperatures shall be measured by means of appropriately located thermocouples and calibrated potentiometers. Insofar as practical, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit. When the temperature of the air is to be measured within a pipe or receiver, the thermocouple shall be exposed to the airstream and shall record the air temperature and not the pipe or receiver temperature.

4.3.9 Rotational speeds. Rotational speeds, measured in revolutions per minute, shall be determined whenever practicable by means of a directly coupled positive counter that will actually count the revolutions for not less than 1 minute. A combination unit that counts the revolutions while measuring the time required in connection with other data is preferred. Strobotachometers may be used provided calibration data can be furnished to attest to their accuracy. The use of hand tachometers will not be acceptable.

4.3.10 Oil consumption. Oil consumption shall be determined by accurately measuring the quantity necessary to maintain a given level in the oil reservoir.

4.3.11 Running-in. The nature and extent of running-in shall be as specified by the manufacturer.

4.3.12 Dewpoint measurements. Dewpoint measurements shall be made following the aftercooler separator and prior to the dehydration equipment and following the dehydration equipment (two readings).

4.4 Preproduction tests.

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4.4.1 Preproduction test sample. One compressor unit shall be subjected to preproduction tests specified in 4.7.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests ..... See 4.5.1

b. Periodic sampling plan and test ..... See 4.5.2

4.5.1 Individual tests. Each compressor unit shall be subjected to the following tests:

a. Examination of product ..... See 4.7.1.

b. Operational tests ..... See 4.7.2.

c. Hydrostatic pressure test .... See 4.7.3.

4.5.1.1 Preinstallation tests. Each spring-loaded relief valve and automatic control device shall be subjected to the appropriate testing under 4.6.

4.5.2 Periodic sampling plans and tests.

4.5.2.1 Compressor sampling plan and test. One compressor unit shall be selected at random from every twenty-five or fraction thereof produced and shall be subjected to a performance demonstration test consisting of 10 percent of all testing hours required in paragraphs 4.7.7.1 thru 4.7.7.1.3. Specified cyclic rates shall remain unchanged and all required data shall be taken.

4.5.2.1.1 Rejection and retest. When a compressor unit selected from a production run fails to meet the specification, compressor units still on hand or produced later shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.1.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of compressor units on hand or produced later shall not be made until it is determined that the compressors meet all the requirements of the specification.

4.5.2.1.1.2 Defects in compressor units already accepted. The investigation of a test failure could indicate that defects may exist in compressor units already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.5.2.2 Shutoff and control valve sampling and test. One of each type (same design, size, and construction) control shutoff valve from each lot (see 6.4.7) of 50 or less shall be selected at random and subjected to the tests specified in 4.7.15.1. In case the valve fails to fulfill all specified test requirements, three additional sample valves from the lot shall be tested in the same manner.

4.5.2.2.1 Shutoff and control valve rejection and retest. If any of the three additional sample valves fail to fulfill all specified test requirements, the entire lot shall be rejected.

4.5.2.3 Burst disc sampling and test. Three burst discs from each lot (see 6.4.7) of 100 or less shall be selected at random and subjected to the test specified in CGA Pamphlet S-1. In case a disc fails at a pressure outside the limits specified, five additional samples from the same lot shall be tested in the same manner.

4.5.2.3.1 Burst disc rejection and retest. If more than one of the three original samples or any of the five additional sample burst discs fails at a pressure outside the limits specified, the entire lot shall be rejected.

4.5.2.4 Check valve sampling and test. One of each type (same design, size and construction) check valve from each lot (see 6.4.7) of 50 or less shall be selected at random and subjected to the tests specified in 4.7.15.2. In case the valve fails to fulfill all specified test requirements, three additional check valves from the lot shall be tested in the same manner.

4.5.2.4.1 Check valve rejection and retest. If any of the three additional sample check valves fail to fulfill all specified test requirements, the entire lot shall be rejected.

4.5.2.5 Automatic control device sampling and test. One of each type (same design, size, and construction) automatic control device from each lot (see 6.4.7) of 50 or less shall be selected at random and subjected to the tests specified in 4.7.15.3. In case the device fails to fulfill all the specified test requirements, three additional samples from the lot shall be tested in the same manner.

4.5.2.5.1 Automatic control device rejection and retest. If any of the three additional samples fail to fulfill all specified test requirements, the entire lot shall be rejected.

4.6 Preinstallation tests.

4.6.1 Spring-loaded relief valve test. Each spring-loaded relief valve shall be subjected to a gradually increasing pressure of clean, dry, oil-free air or nitrogen at its inlet until it opens. The pressure shall then be reduced until the valve completely reseals. This procedure shall be repeated until the valve has opened to discharge, and subsequently reseated, not less than three times. Following this testing, the valve

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shall then be checked for leakage by application of a soap film (film across the outlet and over all outside surfaces) with not less than the maximum working pressure of the component in which the valve will be installed, applied at the inlet. Beginning of discharge below or failure to seal at the pressures required, or indications of leaking during the soap film test, shall be cause for rejection.

4.6.2 Automatic control device test. If the compressor unit is provided with automatic control devices, the devices shall be subjected to not less than 10 full operating cycles under the conditions which they will experience in service. Faulty operation as indicated by failure to control within the limits specified shall be cause for readjustment or rejection. The adequacy of a readjustment operation shall be substantiated by subjecting the device to not less than 25 additional full operating cycles during which the controlled commodity shall be maintained within the specified limits or the device shall be rejected.

4.7 Test methods.

4.7.1 Examination of product. The compressor unit shall be inspected to determine compliance with applicable drawings and the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.7.2 Operational test. The compressor unit shall be operated for 2 hours delivering rated capacity and pressure. Upon completion of the 2-hour period, the compressor unit shall show no signs of malfunctioning, leaks, rough operation, or other irregular operation.

4.7.3 Hydrostatic pressure test. All pressure vessels, including interconnecting piping, operating at pressures in excess of 75 psi shall be hydrostatically tested to not less than 1½ times maximum operating pressures and held at this pressure until examined for leaks and distortion. Leaks or distortion shall be cause for rejection.

4.7.4 Initial teardown inspection. The compressor unit shall first be operated for a break-in period as determined by the manufacturer. The compressor unit shall then be disassembled and all major wearing parts measured and photographed for comparison with those taken after the tests. The photographs and measurements shall be properly identified and included in the test data. After reassembly, the compressor unit shall be serviced with fuels and lubricants, as applicable.

4.7.5 Acceleration test for types II, III, and V compressor units. Prior to conducting any other testing, the compressor unit shall be prepared for shipment. Unless necessary for protection of components from corrosion during testing, preservation for storage will not be required as part of this preparation. The compressor unit shall be subjected to acceleration forces in accordance with MIL-A-8421. After subjection to the 3-g load and prior to the 8-g load, the compressor unit shall be capable of meeting the remainder of the tests without modification, reconditioning, or repair.

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4.7.6 Performance demonstration. The compressor unit shall be started and operated at rated capacity and pressure until temperatures have stabilized. Operation shall then continue for 2 hours with data being recorded every 10 minutes to determine the following:

a. Compressor rotational speed, rpm

b. Discharge pressure, measured at service valve outlet, psi

c. Temperature of air at receiver outlet, degrees Fahrenheit

d. Temperatures of air at the servicing valve, and inlet to the dehydration equipment, when applicable

e. Interstage and aftercooler pressures, psi

f. Free air delivery cfm

g. Intercooler(s) and aftercooler inlet and outlet air temperatures, degrees Fahrenheit

h. Ambient air temperature and pressures, degrees Fahrenheit, inches Hg, wet-bulb temperature, degrees Fahrenheit

i. Lubricating oil temperature, both engine, when applicable, and compressor, degrees Fahrenheit

j. Motor current, voltage, and power factor, or fuel consumption

k. Average temperature of air inside enclosure, when applicable

l. Oil consumption of the compressor, and, when applicable, the engine.

m. When applicable, engine intake manifold temperature and pressure and engine exhaust manifold backpressure

n. Dewpoint measurements for type III compressor units. Measuring instrument to be approved by the procuring activity.

4.7.7 Endurance test.

4.7.7.1 Types II and III compressor units. The compressor unit shall be operated for 1,000 hours as follows:

a. The compressor unit shall be operated for 750 hours delivering not less than rated pressure and capacity. For type II compressor units, 20 continuous hours of intermittent operation (20 cycles) at 250 psi discharge pressure shall be required. Each cycle shall consist of 30 minutes of operation at 250 psi and 30 minutes of operation at 200 psi discharge pressures.

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to both title and number. Copies of this specification and applicable specifica-

b. The compressor unit shall be operated for 100 hours cycling a minimum of 10 times per hour between the cut-in and cutout pressures of the unloader.

c. Type II, class 1 and type III, class 1 compressor units shall be operated for 150 hours for alternate periods of 10 minutes loaded and 50 minutes unloaded. When provided with either a manually or an automatically actuated clutch, fluid coupling or a combination thereof, the coupling shall be actuated a minimum of 10 times per hour throughout the 150-hour period of operation. Type II, classes 2 and 3 and type III, classes 2 and 3 compressor units shall be operated for 150 hours with the controls set on stop-start. The compressor unit shall be required to stop and start approximately 10 times per hour, and where sufficient receiver capacity is available, the stop-start cycle shall be controlled by bleeding sufficient air from the receiver so that the compressor unit must stop and then start the 10 times in an hour.

4.7.7.1.1 Additional data requirements. Data listed under 4.7.6 shall be taken at 1-hour intervals for the 750-hour operation specified in 4.7.7.1. Capacity measurements for the endurance test shall be accurate to ± 5 percent. The amounts of power or fuel, and lubricants consumed shall be recorded for the 1,000 hours. The unloader cycles and the stop-start cycles shall be recorded for 4.7.7.1.1.

4.7.8 Performance demonstration. After completion of endurance testing, the compressor unit shall be resubjected to the testing in 4.7.6.

4.7.9 Final teardown inspection. After completion of the performance demonstration specified in 4.7.8, the compressor unit shall be completely disassembled for inspection. Parts subject to wear or distortion shall be measured or otherwise dimensionally inspected and the measurements compared with similar data compiled prior to the start of the testing for determination of wear or distortion. All parts shall be in an acceptable condition. Wear or distortion that would prevent continued satisfactory operation of the compressor unit shall be cause for considering the part affected as having failed to satisfactorily complete the test.

4.7.9.1.1 Photographs. Photographs shall be made of the corresponding parts which were photographed on the initial teardown inspection (see 4.7.4) and of any other parts which became defective as a result of testing. The photographs shall be properly identified and included in the test data.

4.7.9.2 Nondestruction inspection. Steel parts subject to high stress in operation shall be subjected to magnetic particle inspection (magnaflux) in accordance with MIL-I-6868, and shall exhibit no indications of failure attributable to the testing. Grind checks and subsurface indications of laps, seams, or other forging defects, noted before the test, shall not be cause for rejection. Nonmagnetic castings and forgings shall be subjected to inspection with fluorescent penetrant (black light) in accordance with MIL-I-6866, and shall exhibit no cracks or other defects.

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4.7.10 Static pressure. All gaskets, valves, piping, and seals and fittings shall be tested to 1½ times normal maximum pressure.

4.7.11 Environmental tests. Types II, III, and V compressor units shall be subjected to the following environmental tests:

4.7.11.1 High-temperature exposure. The compressor unit shall be exposed to, but not operated in, an ambient temperature of 160° F for 72 hours. The compressor unit shall then be examined for damage. The ambient temperature shall be reduced to 125° F plus exposure to solar radiation on the surface of 359 BTU/ft/hr and the compressor unit exposed to this condition for 36 hours. While at this temperature, the compressor unit shall be started and operated for 4 hours, delivering rated capacity and pressure. During operation, the data listed under 4.7.6 shall be recorded at 10-minute intervals.

4.7.11.2 Winterized compressor starting. Winterized compressor units shall be exposed to an ambient temperature of -65° F for 36 hours. While exposed to that temperature, the compressor unit shall be heated with a type H-1 heater conforming to MIL-H-4607, or equivalent, in accordance with the manufacturer's instructions, and started. A maximum period of 30 minutes shall be permitted for heating and starting. After starting, the heater shall be disconnected, covers replaced, and the compressor unit operated for 2 hours delivering rated capacity and pressure, and for 2 hours at zero delivery. The data listed under 4.7.6 shall be recorded at 10-minute intervals.

4.7.11.3 Rain test. The enclosed compressor unit shall be exposed to simulated rain in accordance with procedure I of method 506 of MIL-STD-810. After 2 hours and while still being exposed to rain, the compressor unit shall start within 5 minutes and shall continue running for 15 minutes.

4.7.11.4 Wind test. The enclosed compressor unit shall be subjected to a simulated load equivalent to a 70-mph wind. No damage shall result to the doors, hinged covers, or enclosure.

4.7.11.5 Humidity test. The compressor unit or equipment samples representing applicable portions of the compressor unit that might be damaged in accordance with procedure I of method 507 of MIL-STD-810.

4.7.11.6 Fungus test. The compressor unit or equipment samples representing applicable portions of the compressor unit that might be damaged by exposure to fungus attack shall be subjected to the fungus-resistance test specified in procedure I of method 508 of MIL-STD-810.

4.7.11.7 Salt-fog test. The compressor unit or equipment samples representing applicable portions of the compressor unit that might be damaged by exposure to a salt atmosphere shall be subjected to salt-fog in accordance with procedure I of method 509 of MIL-STD-810.

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the supplement thereto. If applicable:

4.7.11.8 Sand and dust test. Enclosed compressor units intende for field use shall be tested in accordance with procedure I, method 510, o MIL-STD-810. Units intended for inside use which do not have an integral enclosure shall be tested by the same procedure using the shipping containe as the enclosure.

4.7.12 Electromagnetic interference test. The compressor uni shall be tested as specified in the detail specification to demonstrate compliance with the applicable requirement of MIL-STD-461.

4.7.13 Mobility tests. In addition to the tests specified in MIL-M-8090, all trailer-mounted compressor units shall be subjected to the following tests:

4.7.13.1 Tilt test. The compressor unit shall be tilted so tha the axle(s) form not less than a 45° angle with the horizontal. While at this position, the lifting mechanism shall be removed, and the compressor unit shall be permitted to fall free. The compressor unit shall be subject to this test from both sides. Overturning of the compressor unit or damage to any part of the compressor unit due to the impact shall be cause for rejection.

4.7.14 Tilted operation. Types II, III, and V trailer-mounte compressor units shall be tilted in four different positions at an angle of 15° from the horizontal. The compressor unit shall be operated for 30 minutes in each of these positions. The compressor unit shall be allowed t stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel systems, after which the compressor unit shall be started and operated for 30 minutes. Fuel or lubrication difficulties or other difficulties due to tilted operation will not be permitted.

4.7.15 Valve and control test.

4.7.15.1 Shutoff and control valve tests. The valve shall be subjected to the following tests in the order specified:

4.7.15.1.1 Cycling test. With the valve outlet plugged or capped, the gas the valve will control shall be applied to its inlet at the maximum working pressure of the component in which the valve will be installed. The valve shall then be subjected to not less than 1,000 cycles of operation (se 6.4.8) without interim valve lubrication, adjustment, or repair. The test shall then be repeated for an additional 1,000 cycles with the outlet open to the atmosphere in which it vents the upstream pressure during the open part of the cycle and the stated working pressure is re-established on the upstream side of the valve during the closed part of the cycle. Damage to or improper operation of the valve as a result of the testing shall be cause for rejection.

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4.7.15.1.2 Overtorque test. At the conclusion of the cycling specified in 4.7.15.1.1, without intervening adjustment or repair, the valve shall be opened, and then closed with a torque of 300 ± 10 pound-inches per inch of nominal size with the valve body at a temperature of not less than 125° F. The test shall be repeated with the valve body at a temperature of -65 ± 10° F. Damage to or improper operation of the valve as a result of the testing shall be cause for rejection.

4.7.15.1.3 Leakage test. Following the overtorque test, and with the valve in the open position and the outlet plugged or capped, the gas the valve will control shall be applied to its inlet at the maximum working pressure of the component in which the valve will be installed, and the valve completely immersed in clear water for not less than 10 minutes. There shall be no evidence of leakage, or the valve shall be considered to have failed the test. This test shall then be repeated with the outlet open and the valve closed with a torque of 60 ± 5 pound-inches per inch of nominal size, and any leakage shall be collected. Total collected gas leakage exceeding 2.5 cubic inches of free gas per hour per half inch of nominal size shall be cause for rejection.

4.7.15.2 Check valve tests. The check valve shall be subjected to the following tests:

4.7.15.2.1 Body leakage test. With the check valve outlet plugged or capped, the gas the valve will control shall be applied to its inlet at the maximum working pressure of the component in which the valve will be installed, and the valve completely immersed in clear water for not less than 10 minutes. There shall be no evidence of leakage.

4.7.15.2.2 Seat leakage test. With the check valve inlet open, the gas the valve will control shall be applied to its outlet at the maximum working pressure the check valve will have to hold in service, the valve completely immersed in clear water, and any leakage collected. Total collected gas leakage exceeding 2.5 cubic inches of free gas per hour per half inch of nominal size shall be considered cause for rejection.

4.7.15.3 Automatic control device test. All automatic control devices, such as, timers, pressure switches, solenoid valves and pneumatic control valves shall be subjected to 2,000 complete operational cycles duplicating that which the devices would normally be expected to perform while operating as part of the compressor unit. Any adjustments necessary shall be performed prior to beginning the test. The test shall be performed in an environment and the controlled medium shall duplicate that of normal operation. In the case of automatically controlled valves, the 2,000-cycle test shall be performed as specified in 4.7.15.1.1. The leakage test specified in 4.7.15.1.3 shall also be performed, except the torque requirement shall not apply. Improper operation of the control devices as a result of testing shall be cause for rejection.

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4.7.16 Audio noise level tests. The criteria specified in 3.8 gives the levels for the operator and for the average at 25 feet for types II, III, and V compressor units. The levels for the operator are the average levels measured at head level (5 feet 4 inches) in the area where the operator is located. This measurement shall be made by slowly moving the measuring microphone in a circle, 2 feet in diameter, centered on the most probable position for an operator. The average level for each octave band shall be determined by a simple arithmetic average of the minimum and maximum noise levels (in decibels) found on this circle. The average levels at 25 feet shall be determined from measurements at head level on a circle 25 feet in radius, centered on the geometrical center of the unit. The noise level for each octave band shall be determined at 12 positions on the circle equally spaced every 30°. The average noise level for each band shall be calculated by a simple arithmetic average of 12 noise levels (in decibles) for the band. Measurements shall be taken with the unit operating at maximum capacity. All acoustic measurements shall be made with equipment in accordance with ASA Bulletins Z24.3 and Z24.10. The flat frequency response (c weighting network) of the sound level meter shall be used in taking the measurements.

4.7.17 Lifting test. The compressor unit shall be subjected to a lifting test to insure compliance with 3.4.14.1 and 3.4.14.2. No breaking, distortion, separation, or loosening of parts shall occur during this test and the compressor unit shall operate satisfactorily upon completion of the test.

4.7.18 Inspection of blowdown system. During the endurance test performance, inspection of the blowdown system shall be conducted to determine compliance with 3.4.15.3.10.

4.7.19 Reliability demonstration. Satiafactory completion of all the tests specified in this specification and in the detail specification without relevant failure damonstrates compliance with,the quantative reliability requirements of this specification.

4.7.19.1 Relevant failure. A failure is the inability of the item to perform any of its intended and specified functions within the specified limits. All failures are considered relevant until classified as not relevant by one of the following provisions and subparagraphs:

a. A secondary or dependent failure is caused by the failure of associated items. For every secondary failure classified as not relevant, a primary or independent failure shall be identified. Only one secondary not-relevant failure of the same part (same drawing number or other unique identifier) shall be allowed during the test. The second and later failures of the same part shall be counted as relevant and corrective action shall be taken to reduce the failure rate of such parts.

b. A test operator or test facility induced failure may be classified as not-relevant if it can be substantiated that the equipment was subjected to operation or stress conditions beyond specified limits.

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c. Changes to the item to correct a deficiency that caused a failure shall not classify such a failure as not-relevant until an item incorporating the change has been demonstrated, by test, to meet the MTBF requirement.

d. Preventive maintenance, servicing and adjustments may be counted as not-relevant if such actions are specified as scheduled normal maintenance in the technical manuals for the equipment.

e. Complete data shall be taken and retained of the analysis, tests, and other actions taken to fully justify each classification of a failure as not-relevant.

f. Any minor malfunction that does not prevent the item's meeting specified performance or does not cause failure of another part need not be counted as a relevant failure unless it is a pattern failure, but all malfunctions shall be described in the test logs and included in any final summary prepared.

g. All failures shall be analyzed to determine if pattern failures (see 6.4.11) exist. The existance of pattern failures shall be cause for rejection until changes are made which have been demonstrated by test to eliminate the pattern failures.

4.7.19.2 Reliability demonstration continuation. If at the completion of the preproduction tests the number of relevant failures exceeds that specified in 4.7.19, the test specified in 4.7.7 shall be continued and shall be considered the reliability test. Measurements and observations shall be continued as time accumulates on the item. An accept decision shall be made when the following inequality holds:

$$\Sigma t \geq \frac{1}{2}\chi^2_{0.1,\ 2r+2} \quad \text{(MTBF)}$$

(Accumulated test operating time)

must be equal to or greater than

½ (Value of Chi squared distribution at 0.1 risk of error and 2r plus 2 degrees of freedom times (the specified MTBF)

Where r is the number of accumulated relevant failures. Note that tables of Chi squared should be used that yield approximately 4.6 at 0.1 risk (0.9 confidence of a correct decision) and no failures (2r plus 2 equals 2).

4.7.19.2.1 Truncation. If a decision can not be reached by the method specified in 4.7.19.2 before the accumulated test operating time is twice the specified endurance test time, the test shall be truncated or terminated at that time unless continued testing is authorized by the procuring activity. If at test termination the inequality still does not hold (the right side is greater), the equipment shall be considered to have failed the test.

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4.7.20 Maintainability. Maintainability demonstration shall comply with MIL-STD-471.

5. PREPARATION FOR DELIVERY. Not applicable to this specification.

6. NOTES

6.1 Intended use. The compressor units covered by this specification are for use by the USAF wherever the need for compressed air arises.

6.2 Items to be considered in detail specification preparation. The following items should be considered, and appropriate requirements inserted as necessary, when an air compressor detail specification is prepared:

a. Type of compressor and class of mobility required (see 1.2)

b. If the number of stages or the maximum compression ratio is to be other than as specified in 3.4.6.1.

c. If a compressor powered by a gasoline engine is not to be provided with a constant-speed type unloader, or if a compressor powered by an electric motor is not to be provided with a combination constant-speed-type unloader and start-stop control (see 3.4.6.2).

d. If a specific range between loading and unloading is required (see 3.4.15.2.8).

e. If type III compressors are not to be equipped with lubrication oil filters (see 3.4.15.3.1.2).

f. If an electrical starting system is not required on a gasoline-powered compressor and specify starting and system voltage (see 3.4.8.1.11).

g. If the generator on the group II engine is to be other than as specified in MIL-E-11275 or MIL-E-11276 (see 3.4.8.1.11.4).

h. If motor other than the squirrel-cage induction motor in accordance with CC-M-641 is to be provided (see 3.4.8.2).

i. If an hourmeter is to be provided (see 3.4.8.4).

j. If other than military lubricants are to be used (see 3.4.9).

k. If a 2-wheel or a 4-wheel trailer is to be used for a trailer-mounted compressor (see 3.4.10.1).

l. Mobility requirements if other than those specified for MIL-M-8090 type II mobility (see 3.4.10.1).

m. If other than a full-swivel landing wheel is required with the 2-wheel mounted compressor (see 3.4.10.1.2).

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n. If multilevel mounting is to be permitted (see 3.4.10.3).

o. If a weatherproofed enclosure is not required (see 3.4.11).

p. If a safety factor of less than 4 to 1 is to be used in the design of pressurized components (see 3.4.5).

q. If the allowable valve leakage rate is to be other than that permitted by 3.4.8.3.2.

r. If the required closing torque is to be other than that permitted by 3.4.8.3.3.

s. If a particular finish is to be specified when more than one finish process is permitted by MIL-STD-808 (see3.9.1).

t. If military standard engines and accessories and/or H-1 heaters are required to be furnished by the Government, and if so, the description and quantity (see 3.13).

6.3 Fire and casualty hazards. The bidder should submit proof to the procuring activity that the electrical components specified in 3.4.7 that he proposes to supply under this specification conform to the standards of the Underwriters' Laboratories, Inc. as regards fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as evidence of conforming to the specification requirements. In lieu of the label or listing, the bidder may submit independent proof satisfactory to the procuring activity that this compressor conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc. Compliance with the fire and casualty hazards requirements does not absolve the bidder from complete compliance with the other requirements of this specification.

6.4 Definitions. For purposes of this specification, the following definitions will apply:

6.4.1 Compressor unit. Compressor unit is defined as the complete air compressor package including prime mover(s), compressor, fans, electrical equipment, receivers and trailer chassis.

6.4.2 Pressure. Unless otherwise specified as absolute or denoted by the symbol psia, all pressures referred to herein will be interpreted as pounds per square inch gage (psig).

6.4.3 Free air. Free air is defined as atmospheric conditions at the compressor unit inlet. Hot airstreams will be effectively shielded from the intake opening.

6.4.4 Dewpoints. All reference dewpoints are at standard atmospheric pressure, unless otherwise specified in the text.

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6.4.5 Ambient temperature. Ambient temperature is defined as the drybulb temperature of the atmospheric air surrounding and in the vicinity of the air compressor and unaffected by heat emanating from the air compressor itself.

6.4.6 Standard air. Standard air is defined as air at 68°F or 528° Rankine under a pressure of 29.92 inch Hg absolute with a relative humidity of 36 percent.

6.4.7 Lot. A lot is defined as a group of identical items manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

6.4.8 Cycle of operation. A cycle of operation will be considered as opening with maximum working pressure at the inlet from a closing torque of 60 ± 5 pound-inches per inch of valve nominal size to not less than the half-open position, and reclosing with a torque of 60 ± 5 pound-inches per inch of valve nominal size.

6.4.9 Reliability. Reliability is the probability that material will perform a specified function without failure under given conditions for a specified period. In general, the instantaneous failure rate is fairly constant throughout the normal operating life of the equipment. Under this condition, the probability of no failure for any particular time period decreases exponentially as a function of the length of the period and of the MTBF. This leads to the following relationship for minimum reliability, R min

$$R_{min} = e^{-t/MTBF}$$

Where t is the time period of mission length

and MTBF is the minimum value demonstrated by test or the lower confidence limit of the MTBF described below.

6.4.10 Mean-time-between-failure. In general, MTBF is the average or arithmetic mean of the operating time between failures or the mean-time-between-failure. As used in this specification, the term MTBF means the minimum MTBF acceptable at the specified confidence. Note that this value calculated from the demonstration data is less than the point estimate of the true MTBF which would be the sum of the test time divided by the sum of the relevant failures. MTBF is often used as an expression for MCBF, mean-cycles-between-failures (where t must then be expressed in cycles to obtain mission reliability). MTBF is also used interchangeable with MTTF, mean-time-to-fail, which applies to items that are expended at first failure and are not repairable. If an exponential distribution of failure is assumed, the Chi squared relationship specified in 4.7.19.2 is appropriate for a lower confidence limit on the estimated true MTBF and therefore, for criteria to determine if the MTBF demonstrated equals or exceeds the MTBF specified as minimum. If no relevant failures are observed,

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at 0.9 confidence this becomes

demonstrated MTBF (min)

$$= \frac{\text{accumulated test operating time}}{2.3}$$

6.4.11 Pattern failures. Pattern failures are defined as two or more failures of the same part in the same or equivalent application during the course of testing whose failure rate exceeds that predicted at random.

6.5 International standardization. Certain provisions of paragraph 3.4.15.3.3.3, 3.4.15.3.8 and 3.4.15.3.3.1 of this specification are subject of international standardization agreement ASCC Air Standard 14/7. Certain provisions of paragraph 3.4.15.3.4 and 3.4.15.3.9 of this specification are the subject of international standardization agreement ASCC Air Standard 11/15.

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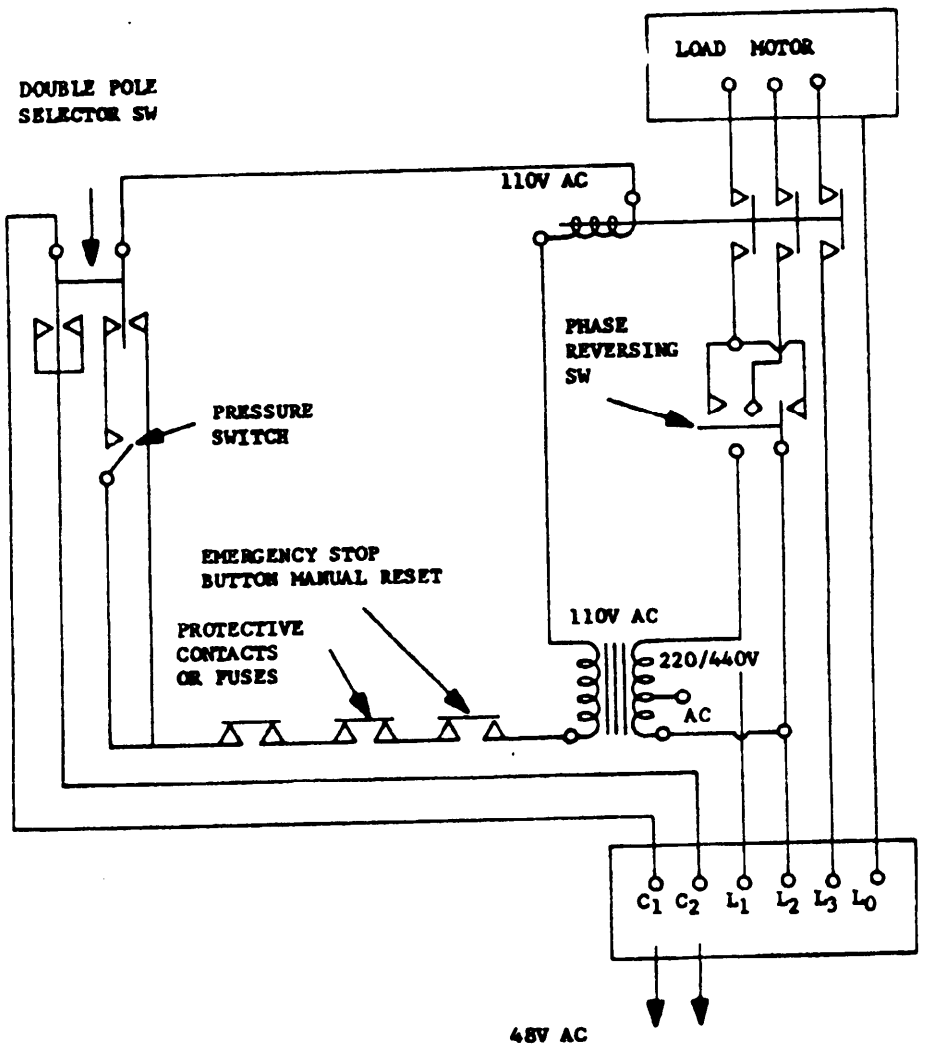


FIGURE 1. WIRING DIAGRAM

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